The Answer to the Exercise on Mathematical Analysis

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Course: Mathematical Analysis (I)

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November 6, 2026

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1 Set theory

Exercise 1.1.
Show that $\varnothing$ is a subset of any set.

Solution.
For any set A, $\varnothing$ is a subset of A if $a \in \varnothing \Rightarrow a \in A$. Since $\varnothing$ has no elements, this condition holds vacuously. Therefore, $\varnothing \subseteq A$. ■

Exercise 1.2.
If A, B, C are three sets, prove that

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

and

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C).$$

Solution.

$$\begin{aligned}(A \cup B) \cap C &= \{x \mid (x \in A \vee x \in B) \wedge x \in C\} \\ &= \{x \mid (x \in A \wedge x \in C) \vee (x \in B \wedge x \in C)\} \\ &= \{x \mid (x \in A \cap C) \vee (x \in B \cap C)\} \\ &= (A \cap C) \cup (B \cap C)\end{aligned}$$

$$\begin{aligned}(A \cap B) \cup C &= \{x \mid (x \in A \wedge x \in B) \vee x \in C\} \\ &= \{x \mid (x \in A \vee x \in C) \wedge (x \in B \vee x \in C)\} \\ &= \{x \mid (x \in A \cup C) \wedge (x \in B \cup C)\} \\ &= (A \cup C) \cap (B \cup C)\end{aligned}$$

■

Exercise 1.3.
If A and B are sets, prove that

$$A \subseteq B \iff A \cap B = A \iff A \cup B = B.$$

Solution.

(1) $A \subset B \implies A \cap B = A$

$\forall x \in A \cap B,\ x \in A \wedge x \in B \implies A \cap B \subset A$

$\because A \subset B \therefore \forall a \in A,\ a \in B \implies a \in A \cap B \implies A \subset A \cap B$

By Axiom of Extension, $A \cap B = A$

(2) $A \cap B = A \implies A \cup B = B$

$\forall x \in A \cup B,\ x \in A \vee x \in B \implies x \in A \cap B \vee x \in B$

$\implies (x \in A \wedge x \in B) \vee x \in B \implies x \in B \implies A \cup B \subset B$

$\forall b \in B,\ b \in A \cup B = \{x \mid x \in A \vee x \in B\}$

By Axiom of Extension, $A \cup B = B$

(3) $A \cup B = B \implies A \subset B$

$\forall a \in A,\ a \in A \cup B \implies a \in B$

Hence, $A \subset B$

■

Exercise 1.4.
Let X be a set. For a subset A of X, we define $A' \equiv A^c := X \setminus A$. If A and B are two subsets of X, prove De Morgan's laws:

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Solution.
We prove both identity relations via equivalent logical statements for element membership.

Part 1: Prove $(A \cup B)^c = A^c \cap B^c$. For any $x \in X$, we have:

$$\begin{aligned} x \in (A \cup B)^c &\iff x \in X \setminus (A \cup B) \\ &\iff x \in X \text{ and } x \notin (A \cup B) \\ &\iff x \in X \text{ and } \neg(x \in A \vee x \in B) \\ &\iff x \in X \text{ and } (x \notin A \wedge x \notin B) \\ &\iff (x \in X \text{ and } x \notin A) \wedge (x \in X \text{ and } x \notin B) \\ &\iff x \in A^c \wedge x \in B^c \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

Hence, $(A \cup B)^c = A^c \cap B^c$.

Part 2: Prove $(A \cap B)^c = A^c \cup B^c$. For any $x \in X$, we have:

$$\begin{aligned} x \in (A \cap B)^c &\iff x \in X \setminus (A \cap B) \\ &\iff x \in X \text{ and } x \notin (A \cap B) \\ &\iff x \in X \text{ and } \neg(x \in A \wedge x \in B) \\ &\iff x \in X \text{ and } (x \notin A \vee x \notin B) \\ &\iff (x \in X \text{ and } x \notin A) \vee (x \in X \text{ and } x \notin B) \\ &\iff x \in A^c \vee x \in B^c \\ &\iff x \in A^c \cup B^c. \end{aligned}$$

Hence, $(A \cap B)^c = A^c \cup B^c$. ■

Exercise 1.5.
Find $\mathcal{P}(\{a, b\})$.

Solution.
By definition of the **power set**, $\mathcal{P}(S)$ is the collection of all subsets of S. The subsets of the set $\{a, b\}$ are $\varnothing$, $\{a\}$, $\{b\}$, and $\{a, b\}$.

Therefore,

$$\mathcal{P}(\{a, b\}) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}.$$

■

Exercise 1.6.
Show that every ordered pair is a set. (Hint: $(a, b) \subseteq \mathcal{P}(\{a, b\})$.)

Solution.
By the definition of ordered pair, an ordered pair is defined as $(a, b) = \{\{a\}, \{a, b\}\}$.

For any sets a and b, the set $\{a, b\}$ exists by the **Axiom of Pairing**. The subsets $\{a\}$ and $\{a, b\}$ are elements of the power set $\mathcal{P}(\{a, b\})$. That is,

$$\{a\} \in \mathcal{P}(\{a, b\}) \quad \text{and} \quad \{a, b\} \in \mathcal{P}(\{a, b\}).$$

Consequently, all elements of (a, b) belong to $\mathcal{P}(\{a, b\})$, which means:

$$(a, b) = \{\{a\}, \{a, b\}\} \subseteq \mathcal{P}(\{a, b\}).$$

By the **Axiom of Specification**, any subset of a set defined by a logical predicate is itself a set. Thus, (a, b) is a set. ■

Exercise 1.7.
If $(a, b) = (c, d)$, show that $a = c$ and $b = d$.

Solution.
By definition, $(a, b) = \{\{a\}, \{a, b\}\}$ and $(c, d) = \{\{c\}, \{c, d\}\}$. The equality $(a, b) = (c, d)$ implies:

$$\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}.$$

We consider two exhaustive cases:

Case 1: $a = b$. In this case, $(a, b) = \{\{a\}, \{a, a\}\} = \{\{a\}\}$. Thus,

$$\{\{c\}, \{c, d\}\} = \{\{a\}\}.$$

This forces $\{c\} = \{a\}$ and $\{c, d\} = \{a\}$, which yields $c = a$ and $d = a = b$. Hence, $a = c$ and $b = d$.

Case 2: $a \neq b$. Since $\{a\} \in \{\{a\}, \{a, b\}\}$, we must have $\{a\} \in \{\{c\}, \{c, d\}\}$, so either $\{a\} = \{c\}$ or $\{a\} = \{c, d\}$.

- If $\{a\} = \{c, d\}$, then $c = d = a$, which implies $(c, d) = \{\{a\}\}$. This contradicts the fact that (a, b) contains two distinct elements $\{a\}$ and $\{a, b\}$.
- Therefore, $\{a\} = \{c\}$, which implies $a = c$.

Substituting $a = c$ into the original equality gives $\{\{a\}, \{a, b\}\} = \{\{a\}, \{a, d\}\}$. Since $\{a, b\} \in \{\{a\}, \{a, d\}\}$ and $\{a, b\} \neq \{a\}$ (because $a \neq b$), it follows that:

$$\{a, b\} = \{a, d\}.$$

Since $b \in \{a, b\}$ and $b \neq a$, we must have $b \in \{a, d\}$ with $b \neq a$, which forces $b = d$.

In both cases, we conclude that $a = c$ and $b = d$. ■

Exercise 1.8.
If A and B are sets, prove that $A \times B$ is a set.

Solution.
The Cartesian product of two sets A and B is defined as

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

For any $a \in A$ and $b \in B$, both elements belong to the union: $a, b \in A \cup B$. This implies $\{a\} \subseteq A \cup B$ and $\{a, b\} \subseteq A \cup B$, so by the definition of the power set,

$$\{a\} \in \mathcal{P}(A \cup B) \quad \text{and} \quad \{a, b\} \in \mathcal{P}(A \cup B).$$

Consequently, the set $(a, b) = \{\{a\}, \{a, b\}\}$ is a subset of $\mathcal{P}(A \cup B)$, which means

$$(a, b) \in \mathcal{P}(\mathcal{P}(A \cup B)).$$

By the Axiom of Union and the Axiom of Power Set, the set $\mathcal{P}(\mathcal{P}(A \cup B))$ exists. Therefore, by the Axiom of Specification,

$$A \times B = \{z \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid \exists a \in A, \exists b \in B \text{ such that } z = (a, b)\}$$

is a well-defined set. ■

Exercise 1.9.
If A, B and X are sets, prove that

(i) $(A \cup B) \times X = (A \times X) \cup (B \times X)$,

(ii) $(A \cap B) \times X = (A \times X) \cap (B \times X)$,

(iii) $(A - B) \times X = (A \times X) - (B \times X)$.

Solution.
We prove each identity by logical equivalence of element membership.

Part (i): Prove $(A \cup B) \times X = (A \times X) \cup (B \times X)$.

$$\begin{aligned}
(A \cup B) \times X &= \{(a, b) \mid a \in A \cup B, b \in X\} \\
&= \{(a, b) \mid (a \in A \vee a \in B) \wedge b \in X\} \\
&= \{(a, b) \mid (a \in A \wedge b \in X) \vee (a \in B \wedge b \in X)\} \\
&= \{(a, b) \mid a \in A, b \in X\} \cup \{(a, b) \mid a \in B, b \in X\} \\
&= (A \times X) \cup (B \times X)
\end{aligned}$$

Part (ii): Prove $(A \cap B) \times X = (A \times X) \cap (B \times X)$.

$$\begin{aligned}
(A \cap B) \times X &= \{(a, b) \mid a \in A \cap B, b \in X\} \\
&= \{(a, b) \mid (a \in A \wedge a \in B) \wedge b \in X\} \\
&= \{(a, b) \mid (a \in A \wedge b \in X) \wedge (a \in B \wedge b \in X)\} \\
&= \{(a, b) \mid a \in A, b \in X\} \cap \{(a, b) \mid a \in B, b \in X\} \\
&= (A \times X) \cap (B \times X)
\end{aligned}$$

Part (iii): Prove $(A - B) \times X = (A \times X) - (B \times X)$.

$$\begin{aligned}
(A - B) \times X &= \{(a, b) \mid a \in A - B, b \in X\} \\
&= \{(a, b) \mid a \in A \wedge \neg(a \in B) \wedge b \in X\} \\
&= \{(a, b) \mid a \in A \wedge b \in X \wedge [\neg(a \in B) \vee \neg(b \in X)]\} \\
&= \{(a, b) \mid a \in A \wedge b \in X \wedge \neg(a \in B \wedge b \in X)\} \\
&= (A \times X) - (B \times X)
\end{aligned}$$

■

Exercise 1.10.
If A and B are sets, prove that $A \times B = \varnothing \iff A = \varnothing$ or $B = \varnothing$.

Solution.
We prove both directions of the equivalence.

($\Leftarrow$) Assume $A = \varnothing$ or $B = \varnothing$. If $A = \varnothing$, then there exists no element $x \in A$. Consequently, no ordered pair (x, y) with $x \in A$ and $y \in B$ can be formed. Thus, $A \times B = \varnothing$. Similarly, if $B = \varnothing$, no element $y \in B$ exists, yielding $A \times B = \varnothing$.

($\Rightarrow$) We prove the contrapositive: if $A \neq \varnothing$ and $B \neq \varnothing$, then $A \times B \neq \varnothing$. Since $A \neq \varnothing$, there exists at least one element $x \in A$. Since $B \neq \varnothing$, there exists at least one element $y \in B$. Thus, the ordered pair (x, y) satisfies $x \in A$ and $y \in B$, which implies $(x, y) \in A \times B$. Hence, $A \times B \neq \varnothing$.

Therefore, $A \times B = \varnothing \iff A = \varnothing$ or $B = \varnothing$. ■

Exercise 1.11.
Show that $\varnothing$ is a relation.

Solution.
By definition, a binary relation R from a set X to a set Y is any subset of the Cartesian product $X \times Y$.

Since the empty set $\varnothing$ is a subset of every set, we have:

$$\varnothing \subseteq X \times Y$$

for any choice of sets X and Y. Therefore, $\varnothing$ satisfies the formal definition of a relation.
■

Exercise 1.12.
If A and B are sets, prove that $A \times B$ is a relation.

Solution.
By definition, a binary relation from a set A to a set B is any subset of the Cartesian product $A \times B$.

Since $A \times B \subseteq A \times B$ holds trivially for any set $A \times B$, it follows that $A \times B$ is a relation from A to B. ■

Exercise 1.13.
If R is a relation, $A := \text{domain}(R)$, and $B := \text{range}(R)$, prove that $R \subseteq A \times B$.

Solution.
By definition, the domain and range of a relation R are given by

$$\text{domain}(R) = \{x \mid \exists y \text{ such that } (x, y) \in R\} = A,$$

$$\text{range}(R) = \{y \mid \exists x \text{ such that } (x, y) \in R\} = B.$$

Let $z \in R$. Since R is a relation, z is an ordered pair, so $z = (x, y)$ for some x and y. By definition of domain, $x \in \text{domain}(R) = A$. By definition of range, $y \in \text{range}(R) = B$.

Thus, $(x, y) \in A \times B$, which shows that $R \subseteq A \times B$. ■

Exercise 1.14.
Let X be a set. Prove that $R := \{(A, B) \mid A, B \subseteq X\}$ is an equivalence relation on $\mathcal{P}(X)$.

Solution.
To prove that R is an equivalence relation on the power set $\mathcal{P}(X)$, we must verify three properties:

1. **Reflexivity:** For any $A \in \mathcal{P}(X)$, we have $A \subseteq X$. Thus, $(A, A) \in R$.
2. **Symmetry:** Let $(A, B) \in R$. By definition of R, both $A \subseteq X$ and $B \subseteq X$. This implies $(B, A) \in R$.
3. **Transitivity:** Let $(A, B) \in R$ and $(B, C) \in R$. Then $A \subseteq X$, $B \subseteq X$, and $C \subseteq X$. In particular, $A \subseteq X$ and $C \subseteq X$, so $(A, C) \in R$.

Since R is reflexive, symmetric, and transitive on $\mathcal{P}(X)$, R is an equivalence relation. ■

Exercise 1.15.
Prove that $\varnothing$ is a function.

Solution.
A set f is defined to be a function if f is a relation (i.e., a set of ordered pairs) satisfying the single-valued condition:

$$\forall x, y_1, y_2, \quad ((x, y_1) \in f \wedge (x, y_2) \in f) \implies y_1 = y_2.$$

1. The empty set $\varnothing$ is a relation because $\varnothing \subseteq \varnothing \times \varnothing$. 2. The conditional requirement holds vacuously because there exist no elements $(x, y_1) \in \varnothing$.

Therefore, $\varnothing$ is a function. ■

Exercise 1.16.
Prove that $\varnothing$ is a 1-1 function.

Solution.
A function f is defined to be 1-1 (injective) if it satisfies:

$$\forall x_1, x_2, y, \quad ((x_1, y) \in f \wedge (x_2, y) \in f) \implies x_1 = x_2.$$

Since $\varnothing$ contains no elements, the premise $(x_1, y) \in \varnothing$ is false for all x_1, y. Thus, the implication holds vacuously.

Hence, $\varnothing$ is a 1-1 function. ■

Exercise 1.17.
Prove that a function F is 1-1 if and only if F^{-1} is also a function.

Solution.
Let F be a function. The inverse relation is defined by

$$F^{-1} = \{(y, x) \mid (x, y) \in F\}.$$

We prove both directions of the equivalence:

($\Rightarrow$) Assume F is 1-1. We must show that F^{-1} is a function. Suppose $(y, x_1) \in F^{-1}$ and $(y, x_2) \in F^{-1}$. By definition of F^{-1}, this means $(x_1, y) \in F$ and $(x_2, y) \in F$. Since F is 1-1, $(x_1, y) \in F \wedge (x_2, y) \in F \implies x_1 = x_2$. Hence, F^{-1} satisfies the single-valued property and is a function.

($\Leftarrow$) Assume F^{-1} is a function. We must show that F is 1-1. Suppose $(x_1, y) \in F$ and $(x_2, y) \in F$. By definition of F^{-1}, we have $(y, x_1) \in F^{-1}$ and $(y, x_2) \in F^{-1}$. Since F^{-1} is a function, it must map y to a unique value, so $x_1 = x_2$. Hence, F is 1-1.

Therefore, F is 1-1 if and only if F^{-1} is a function. ■

Exercise 1.18.
Suppose $f : X \to Y$ is a function. Show that the following are equivalent:

(i) f is 1-1,

(ii) $f(A \setminus B) = f(A) \setminus f(B)$ for all subsets A and B of X,

(iii) $f(A \cap B) = f(A) \cap f(B)$ for all subsets A and B of X.

Solution.
We establish equivalence by proving $(i) \implies (ii) \implies (iii) \implies (i)$.

Step 1: $(i) \implies (ii)$**.**
Assume f is 1-1. Let $A, B \subseteq X$.

- ($\subseteq$) Let $y \in f(A \setminus B)$. Then $y = f(x)$ for some $x \in A \setminus B$, meaning $x \in A$ and $x \notin B$. Since $x \in A$, $y = f(x) \in f(A)$. Suppose for contradiction that $y \in f(B)$. Then $y = f(x')$ for some $x' \in B$. Since f is 1-1, $f(x) = f(x') \implies x = x' \in B$, which contradicts $x \notin B$. Thus $y \notin f(B)$, so $y \in f(A) \setminus f(B)$.

- ($\supseteq$) Let $y \in f(A) \setminus f(B)$. Then $y \in f(A)$ and $y \notin f(B)$. Since $y \in f(A)$, $y = f(x)$ for some $x \in A$. If $x \in B$, then $y = f(x) \in f(B)$, a contradiction. Hence $x \notin B$, so $x \in A \setminus B$. Therefore $y = f(x) \in f(A \setminus B)$.

Thus $f(A \setminus B) = f(A) \setminus f(B)$.

Step 2: $(ii) \implies (iii)$**.**
Assume (ii) holds. For any subsets $A, B \subseteq X$, notice that $A \cap B = A \setminus (A \setminus B)$. Using property (ii) twice:

$$f(A \cap B) = f(A \setminus (A \setminus B)) = f(A) \setminus f(A \setminus B) = f(A) \setminus (f(A) \setminus f(B)) = f(A) \cap f(B).$$

Thus $f(A \cap B) = f(A) \cap f(B)$.

Step 3: $(iii) \implies (i)$**.**
Assume (iii) holds. Let $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$. Consider singletons $A = \{x_1\}$ and $B = \{x_2\}$. Then $f(A) = \{f(x_1)\}$ and $f(B) = \{f(x_2)\} = \{f(x_1)\}$. By property (iii),

$$f(\{x_1\} \cap \{x_2\}) = f(\{x_1\}) \cap f(\{x_2\}) = \{f(x_1)\} \neq \varnothing.$$

This implies $\{x_1\} \cap \{x_2\} \neq \varnothing$, which means $x_1 = x_2$. Hence, f is 1-1.

Since $(i) \implies (ii) \implies (iii) \implies (i)$, the statements are equivalent. ■

Exercise 1.19.
Suppose $F : X \to Y$ is a function. Let $\{A_i\}_{i \in I}$ be a family of subsets of Y, and let $\{B_j\}_{j \in J}$ be a family of subsets of X. Prove that

(a) $F\left(\bigcup_{j \in J} B_j\right) = \bigcup_{j \in J} F(B_j)$,

(b) $F^{-1}\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} F^{-1}(A_i)$,

(c) $F^{-1}\left(\bigcap_{i \in I} A_i\right) = \bigcap_{i \in I} F^{-1}(A_i)$,

(d) $F^{-1}(A^c) = [F^{-1}(A)]^c$, where $A \subseteq Y$.

Solution.
We prove each equality via logical equivalence of element membership.

Part (a): For any $y \in Y$,

$$\begin{aligned} y \in F\left(\bigcup_{j \in J} B_j\right) &\iff \exists x \in \bigcup_{j \in J} B_j \text{ such that } F(x) = y \\ &\iff \exists x \in X, \exists j_0 \in J \text{ s.t. } (x \in B_{j_0} \wedge F(x) = y) \\ &\iff \exists j_0 \in J,\ y \in F(B_{j_0}) \\ &\iff y \in \bigcup_{j \in J} F(B_j). \end{aligned}$$

Thus, $F\left(\bigcup_{j \in J} B_j\right) = \bigcup_{j \in J} F(B_j)$.

Part (b): For any $x \in X$,

$$\begin{aligned} x \in F^{-1}\left(\bigcup_{i\in I} A_i\right) &\iff F(x) \in \bigcup_{i\in I} A_i \\ &\iff \exists i_0 \in I \text{ such that } F(x) \in A_{i_0} \\ &\iff \exists i_0 \in I \text{ such that } x \in F^{-1}(A_{i_0}) \\ &\iff x \in \bigcup_{i\in I} F^{-1}(A_i). \end{aligned}$$

Thus, $F^{-1}\left(\bigcup_{i\in I} A_i\right) = \bigcup_{i\in I} F^{-1}(A_i)$.

Part (c): For any $x \in X$,

$$\begin{aligned} x \in F^{-1}\left(\bigcap_{i\in I} A_i\right) &\iff F(x) \in \bigcap_{i\in I} A_i \\ &\iff \forall i \in I,\ F(x) \in A_i \\ &\iff \forall i \in I,\ x \in F^{-1}(A_i) \\ &\iff x \in \bigcap_{i\in I} F^{-1}(A_i). \end{aligned}$$

Thus, $F^{-1}\left(\bigcap_{i\in I} A_i\right) = \bigcap_{i\in I} F^{-1}(A_i)$.

Part (d): For any $x \in X$ and subset $A \subseteq Y$,

$$\begin{aligned} x \in F^{-1}(A^c) &\iff F(x) \in A^c \\ &\iff F(x) \in Y \setminus A \\ &\iff F(x) \notin A \\ &\iff \neg(x \in F^{-1}(A)) \\ &\iff x \in [F^{-1}(A)]^c. \end{aligned}$$

Thus, $F^{-1}(A^c) = [F^{-1}(A)]^c$. ■

Exercise 1.20.
Prove that the symbol "$\sim$" in Definition 1.15 satisfies three conditions in the definition of equivalence relation.

Solution.
The symbol $\sim$ denotes **equipotence** (or equal cardinality) between two sets, meaning $A \sim B$ if and only if there exists a **bijection** $f : A \to B$. We verify the three properties of an equivalence relation:

1. **Reflexivity:** For any set A, the identity function $I_A : A \to A$ defined by $I_A(x) = x$ is a bijection. Therefore, $A \sim A$.

2. **Symmetry:** Suppose $A \sim B$. Then there exists a bijection $f : A \to B$. Since f is bijective, its inverse function $f^{-1} : B \to A$ is also a bijection. Therefore, $B \sim A$.

3. **Transitivity:** Suppose $A \sim B$ and $B \sim C$. There exist bijections $f : A \to B$ and $g : B \to C$. The composition of two bijections, $g \circ f : A \to C$, is also a bijection. Therefore, $A \sim C$.

Hence, $\sim$ is an equivalence relation on any set of sets. ■

Exercise 1.21.
If A is a set, show that the cardinality exists.

Solution.
By the **Well-Ordering Theorem** (which is logically equivalent to the **Axiom of Choice**), every set A can be well-ordered.

Every well-ordered set is order-isomorphic to a unique **ordinal number**. Let Ω be the class of all ordinals that are equipotent to A. Since A can be well-ordered, Ω is non-empty.

The class of ordinal numbers is itself well-ordered by set membership ($\in$). Therefore, the non-empty class Ω contains a least element. This least ordinal, often called the **initial ordinal** equipotent to A, is defined as the cardinality of A, denoted as $\text{card}(A)$ or $|A|$. Thus, the cardinality of any set A exists. ■

Exercise 1.22.
If there exists a 1-1 function $f : A \to B$, show that $\text{card}A \leq \text{card}B$.

Solution.
By definition, the inequality of cardinal numbers $\text{card}A \leq \text{card}B$ is strictly defined by the existence of an **injective** (1-1) function from A to B.

Since we are given a 1-1 function $f : A \to B$, the condition for cardinality inequality is directly satisfied. The set A is equipotent to $f(A)$, which is a subset of B. Hence, $\text{card}A \leq \text{card}B$. ■

Exercise 1.23.
Show that $\text{card}A < \text{card}2^A$ for any set A.

Solution.
This is a proof of **Cantor's Theorem**.

First, we show $\text{card}A \leq \text{card}2^A$. Define a function $f : A \to 2^A$ by $f(x) = \{x\}$. For any $x, y \in A$, if $f(x) = f(y)$, then $\{x\} = \{y\}$, implying $x = y$. Thus, f is 1-1, which establishes $\text{card}A \leq \text{card}2^A$.

Second, we prove $\text{card}A \neq \text{card}2^A$ by contradiction. Suppose there exists a **surjection** $g : A \to 2^A$. Construct the subset $S \subseteq A$ defined by:

$$S = \{x \in A \mid x \notin g(x)\}$$

Since $S \in 2^A$ and g is surjective, there must exist some $y \in A$ such that $g(y) = S$. Consider whether $y \in S$:

- If $y \in S$, then by definition of S, $y \notin g(y)$. Since $g(y) = S$, this implies $y \notin S$, a contradiction.
- If $y \notin S$, then $y \notin g(y)$. By definition of S, this implies $y \in S$, a contradiction.

Since both cases yield logical contradictions, no such surjection g can exist. Consequently, no bijection can exist between A and 2^A.

Therefore, card$A <$ card2^A. ■

Exercise 1.24.
Prove that $\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$ is countable.

Solution.
A set is **countable** if it is equipotent to the set of natural numbers $\mathbb{N}$. Let $\mathbb{N} = \{0, 1, 2, 3, \dots\}$. We construct a function $f : \mathbb{Z} \to \mathbb{N}$ as follows:

$$f(n) = \begin{cases} 2n, & \text{if } n \geq 0 \\ -2n - 1, & \text{if } n < 0 \end{cases}$$

To show f is a bijection:

- **Injectivity:** If $f(a) = f(b)$, they must have the same parity. If both are even, $2a = 2b \implies a = b$. If both are odd, $-2a - 1 = -2b - 1 \implies a = b$. Thus f is 1-1.
- **Surjectivity:** Let $m \in \mathbb{N}$. If m is even, let $n = m/2 \geq 0$; then $f(n) = m$. If m is odd, let $n = -(m + 1)/2 < 0$; then $f(n) = m$. Thus f is onto.

Since f is a bijection, $\mathbb{Z} \sim \mathbb{N}$. Therefore, $\mathbb{Z}$ is countable. ■

Exercise 1.25.
Let A be a nonempty set and let $a \in A$. If $A \sim A \setminus \{a\}$, show A is infinite.

Solution.
By definition, a set is **Dedekind-infinite** if it is equipotent to a proper subset of itself.

Since $a \in A$, the set $A \setminus \{a\}$ is a proper subset of A (as $a \notin A \setminus \{a\}$). We are given that $A \sim A \setminus \{a\}$, meaning there exists a bijection between A and a proper subset of A.

Any set that is Dedekind-infinite is an infinite set (it cannot be in bijection with any finite ordinal). Therefore, A is infinite. ■

Exercise 1.26.
Let α be an ordinal number with $\alpha > 0$ and let S be a nonempty subset of α. Show that S contains the minimum.

Solution.
Let $\beta = \{x \in \alpha \mid x < y \text{ for all } y \in S\}$. The set β consists of all strict lower bounds of S within α. Since α is an ordinal, it is a **transitive set** well-ordered by $\in$. The condition defines β as an initial segment of α, meaning β is itself an ordinal.

Because $S \subseteq \alpha$, we have $\beta \subseteq \alpha$. Since S is nonempty, $\beta \neq \alpha$ (otherwise α would be strictly less than elements in $S \subseteq \alpha$, leading to $y < y$). Since β is an ordinal properly contained in α, we have $\beta \in \alpha$.

We claim $\beta = \min(S)$. Assume $\beta \notin S$. By definition, β is the set of all $x \in \alpha$ strictly less than every $y \in S$. Since $\beta \notin S$, and α is totally ordered, β must be strictly less than every $y \in S$. But this satisfies the defining condition of β, implying $\beta \in \beta$. This violates the **Axiom of Foundation** (no set is an element of itself).

Hence, our assumption is false, and $\beta \in S$. Since $\beta \in S$ and β is defined such that it is less than or equal to every $y \in S$, β is the minimum element of S. ■

Exercise 1.27.
Let α be an ordinal number with $\alpha > 0$ and let S be a subset of α. Suppose that (i) $0 \in S$. (ii) If $y \in \alpha$ is such that $y \subset S$, then $y \in S$. Show that $S = \alpha$.

Solution.
We proceed by **Transfinite Induction**. Assume for contradiction that $S \neq \alpha$. Since S is a subset of α, the relative complement $\alpha \setminus S$ is a nonempty subset of α.

By the well-ordering of ordinals (established in the previous exercise), the nonempty set $\alpha \setminus S$ must contain a minimum element. Let $\gamma = \min(\alpha \setminus S)$.

Since γ is the minimum of $\alpha \setminus S$, any ordinal $x < \gamma$ must satisfy $x \notin \alpha \setminus S$. Because $x \in \alpha$, this implies $x \in S$. Recall that for ordinals, $x < \gamma \iff x \in \gamma$. Thus, for all $x \in \gamma$, we have $x \in S$. This is equivalently written as $\gamma \subseteq S$.

If $\gamma = 0$, then $\gamma \in S$ by property (i), contradicting $\gamma \in \alpha \setminus S$. If $\gamma > 0$, since $\gamma \in \alpha$ and $\gamma \subseteq S$, we note that for ordinals, subsets that are initial segments are elements. More precisely, property (ii) states that if $y \in \alpha$ and $y \subset S$ (where $\subset$ is interpreted as subset inclusion), then $y \in S$. Since $\gamma \subseteq S$, applying property (ii) yields $\gamma \in S$.

This directly contradicts the initial assumption that $\gamma \in \alpha \setminus S$. Hence, $\alpha \setminus S$ must be empty, meaning $S = \alpha$. ■

Exercise 1.28.
Let A_i be sets for each $i \in I$, where I is a set. Show that the family $\{A_i\}_{i \in I}$ is also a set.

Solution.
By the **Axiom Schema of Replacement**, if the domain I is a set, and there exists a functional relation $f(i) = A_i$ mapping each element of I to a unique set A_i, then the range of this mapping is also a set. Thus, the collection $\mathcal{A} = \{A_i \mid i \in I\}$ is a valid set.

Following the provided hint (Axiom of Unions and Axiom of Specification), we can formally define the indexed family as a function (a set of ordered pairs). Let $U = \bigcup_{i \in I} A_i$

(which requires the **Axiom of Unions** operating on the range set $\mathcal{A}$). The Cartesian product $I \times \mathcal{A}$ exists.

By the **Axiom of Specification**, we can construct the set:

$$F = \{(i, X) \in I \times \mathcal{A} \mid X = A_i\}$$

This set F represents the formal family $\{A_i\}_{i \in I}$ as a function with domain I. Because it is separated from a valid bounding set $I \times \mathcal{A}$, it is rigorously proven to be a set. ■

2 Real and Complex Number Systems